

✓ AI Content Calendar Generator

Project 3 — AI Automation Portfolio

This project uses Python, Prompt Engineering, and the OpenAI API to generate structured social media content calendars.

The AI creates content ideas based on:

- Business or brand information
- Industry or niche
- Target audience
- Marketing goals
- Social media platform
- Content duration

Project Goal

The goal of this project is to automate the content-planning process by converting business and marketing requirements into a clear, organized, and actionable social media content calendar.

✓ Create Marketing Inputs

Prompt Design and Core Logic

In this section, we define the business and marketing information required to generate a structured social media content calendar.

✓ Adding the Business and Marketing Variables

```
# Business and marketing information

business_name = "EcoStyle"

industry = "Sustainable fashion"

target_audience = "Environmentally conscious young adults aged 18 to 35"
```

```
marketing_goal = "Increase brand awareness and encourage website visits"

platform = "Instagram"

duration = 7
```

✓ Create the AI Content Calendar Prompt

```
# Create a structured prompt for the AI content calendar

prompt = f"""
You are an expert social media strategist and digital marketing specialist.

Your task is to create a practical, engaging, and platform-specific
content calendar based on the business information below.

Business Name: {business_name}

Industry: {industry}

Target Audience: {target_audience}

Marketing Goal: {marketing_goal}

Social Media Platform: {platform}

Content Calendar Duration: {duration} days

Follow these instructions:

1. Create one content idea for each day.
2. Make every idea relevant to the business and target audience.
3. Align the content with the marketing goal.
4. Use content formats that work well on the selected platform.
5. Include a variety of content types.
6. Avoid repeating the same content idea.
7. Balance educational, engaging, promotional, and brand-building content.
8. Include a clear call to action for each post.
9. Suggest relevant hashtags.
10. Keep all ideas practical and realistic.

Use this output structure:

## Content Strategy Overview

## {duration}-Day Content Calendar

| Day | Content Topic | Content Format | Content Idea | Call to Action | Hashta
```

```
|---|---|---|---|---|---|
"""
```

```
print(prompt)
```

You are an expert social media strategist and digital marketing specialist.

Your task is to create a practical, engaging, and platform-specific content calendar based on the business information below.

Business Name: EcoStyle

Industry: Sustainable fashion

Target Audience: Environmentally conscious young adults aged 18 to 35

Marketing Goal: Increase brand awareness and encourage website visits

Social Media Platform: Instagram

Content Calendar Duration: 7 days

Follow these instructions:

1. Create one content idea for each day.
2. Make every idea relevant to the business and target audience.
3. Align the content with the marketing goal.
4. Use content formats that work well on the selected platform.
5. Include a variety of content types.
6. Avoid repeating the same content idea.
7. Balance educational, engaging, promotional, and brand-building content.
8. Include a clear call to action for each post.
9. Suggest relevant hashtags.
10. Keep all ideas practical and realistic.

Use this output structure:

```
## Content Strategy Overview
```

```
## 7-Day Content Calendar
```

```
| Day | Content Topic | Content Format | Content Idea | Call to Action | Hashtag
|---|---|---|---|---|---|
```

✓ Generate the AI Content Calendar

Step 2.2: AI-Powered Content Calendar Generation

In this section, the structured marketing prompt is sent to the OpenAI API. The AI analyzes the business information and generates a practical, platform-specific content calendar.

```
!pip install -q openai
```

```
# Loading the API key
from google.colab import userdata

api_key = userdata.get("OPENAI_API_KEY")
```

```
# Creating the OpenAI Client
from openai import OpenAI

client = OpenAI(api_key=api_key)
```

Generate the Content Calendar

```
response = client.responses.create(
    model="gpt-5-mini",
    input=prompt
)

content_calendar = response.output_text

print(content_calendar)
```

The expected output may look similar to this:

Content Strategy Overview

The content strategy focuses on increasing EcoStyle's brand awareness by combining sustainable fashion education, behind-the-scenes content, styling tips, community engagement, and product-focused posts.

7-Day Content Calendar

Day	Content Topic	Content Format	Content Idea	
Day 1	Brand Story	Instagram Reel	Introduce EcoStyle's mission and sustainable values	Follow for
Day 2	Sustainable Materials	Carousel	Explain how recycled materials are used	Save this p
Day 3	Styling Tips	Reel	Show three ways to style one sustainable outfit	Share with

Day	Content Topic	Content Format	Content Idea	
Day 4	Behind the Scenes	Instagram Story	Show part of the product-design process	Ask a ques
Day 5	Customer Feature	Image Post	Highlight a customer wearing an EcoStyle product	Tag us in y
Day 6	Sustainability Tips	Carousel	Share practical tips for building a sustainable wardrobe	Save and s
Day 7	Product Promotion	Reel	Feature a popular EcoStyle product	Visit the w

Let's Take a View of the Quality Check

Analyse the output:

- The calendar contains 7 content ideas.
- Each idea is relevant to sustainable fashion.
- The target audience is considered.
- The ideas support brand awareness and website visits.
- The formats are suitable for Instagram.
- The content ideas are varied.
- Each post includes a call to action.
- Relevant hashtags are included.
- The calendar is clear and easy to use.

✓ Step 3.1: Test a New Business Scenario

We'll create a content calendar for an e-commerce skincare brand on TikTok.

- Part A — Replace the Marketing Inputs

Let's come back to our existing business-input code cell and replace the EcoStyle information with:

```
# Business and marketing information

business_name = "GlowNaturals"

industry = "Natural skincare and beauty products"

target_audience = """
Young adults aged 18 to 35 who are interested in affordable,
natural, and beginner-friendly skincare products
"""

marketing_goal = """
Increase product awareness, build trust with potential customers,
and encourage visits to the online store
"""
```

```
"""  
  
platform = "TikTok"  
  
duration = 7
```

The AI should generate a TikTok-focused content calendar with ideas such as:

- Skincare routines
- Product demonstrations
- Educational skincare tips
- Ingredient explanations
- Common skincare mistakes
- Behind-the-scenes content
- Customer-focused content
- Product promotion

Evaluate the Output

Review the generated calendar using this checklist:



Evaluation Area	Expected Result
Business relevance	Content relates to natural skincare
Audience relevance	Suitable for young adults and skincare beginners
Marketing alignment	Supports awareness, trust, and online-store visits
Platform suitability	Uses TikTok-friendly formats
Content variety	Includes educational, engaging, and promotional ideas
Practicality	Ideas are realistic to create
Calls to action	Every post has a useful CTA
Hashtags	Hashtags are relevant and not repetitive
Write Pass or Needs Improvement beside each area.	

✓ Refine the Prompt

Now we'll improve the prompt so the generated calendar is more strategic, practical, and consistent.

Replacing our current prompt with:

```
# Refined prompt for AI content calendar generation

prompt = f"""
You are an expert social media strategist, digital marketer,
and content planner.

Your task is to create a practical, strategic, engaging, and
platform-specific content calendar using the business information below.

Business Name:
{business_name}

Industry:
{industry}

Target Audience:
{target_audience}
```

Primary Marketing Goal:
{marketing_goal}

Social Media Platform:
{platform}

Content Calendar Duration:
{duration} days

Follow these rules carefully:

1. Create exactly one content idea for each day.
2. Make every idea relevant to the business, industry, and target audience.
3. Align every post with the primary marketing goal.
4. Use content formats that are suitable for {platform}.
5. Include a balanced mix of:
 - Educational content
 - Engaging or interactive content
 - Trust-building content
 - Brand-awareness content
 - Product or service promotion
6. Avoid repeating the same topic, format, or content angle.
7. Make the ideas practical for a small business or beginner content creator.
8. Include a clear and specific call to action for every post.
9. Suggest 3 to 5 relevant hashtags for every post.
10. Avoid generic ideas that could apply to any business.
11. Do not invent product features, discounts, customer results, or business facts that were not provided.
12. Keep each calendar entry concise and actionable.

First, provide a short content strategy overview that explains:

- The main content themes
- How the content supports the marketing goal
- How the content is adapted for {platform}

Then create the calendar using exactly this format:

Content Strategy Overview

{duration}-Day Content Calendar

Day	Content Pillar	Topic	Format	Content Idea	Call to Action	Hasht
---	---	---	---	---	---	---

"""

The refined prompt adds:

- Content pillars to make the strategy more organized

- A clearer balance between education, engagement, trust, awareness, and promotion
- Platform-specific requirements
- Practicality for small businesses
- A rule against generic ideas
- A rule against inventing business claims
- More specific hashtag requirements
- A strategic overview before the calendar

✓ Step 3.2: Interactive Content Calendar Generator

This section allows users to enter their business and marketing information and generate a customized AI-powered content calendar.

```
# Collect business and marketing information from the user

business_name = input("Enter the business or brand name: ")

industry = input("Enter the industry or niche: ")

target_audience = input("Describe the target audience: ")

marketing_goal = input("Enter the primary marketing goal: ")

platform = input("Enter the social media platform: ")

duration = int(input("Enter the content calendar duration in days: "))

print("\nBusiness and marketing information received successfully.")
```

Enter the business or brand name:

TechStart Academy

Enter the industry or niche:

Online technology education

Describe the target audience:

Students and beginners aged 18 to 30 who want to learn practical technology and AI skills

Enter the primary marketing goal:

Increase course awareness, build trust, and encourage course registrations

Enter the social media platform:

LinkedIn

Enter the content calendar duration in days:

7

After entering the final value see:

Business and marketing information received successfully.

The calendar may include:

- AI or technology learning tips
- Beginner-friendly educational posts
- Career-development content
- Student-focused engagement questions
- Course-related content
- Trust-building educational content
- Clear registration calls to action

The Next output may different.

Final Test with different platforms



```
# AI Content Calendar Generator

## Project Goal

## Step 2: Prompt Design and Core Logic
- Business and marketing variables
- Structured AI prompt

## Step 2.2: AI-Powered Content Calendar
Generation
- OpenAI API connection
- First generated content calendar

## Step 3.1: Testing and Prompt Refinement
- GlowNaturals TikTok scenario
```

AI Content Calendar Generator

Project Goal

Step 2: Prompt Design and Core Logic

- Business and marketing variables

```
- Refined marketing prompt

## Step 3.2: Interactive Content Calendar
Generator
- User input workflow

## Sample Output : TikTok Content Calendar
for GlowNaturals
- Generated content calendar

## Sample Output 2: LinkedIn Content
Calendar for TechStart Academy
- Generated content calendar

## Next Sample Output 3: Instagram Content
Calendar for UrbanCraft Store
- Generated content calendar
```

- Structured AI prompt

Step 2.2: AI-Powered Content Calendar Generation

- OpenAI API connection
- First generated content calendar

Step 3.1: Testing and Prompt Refinement

- GlowNaturals TikTok scenario
- Refined marketing prompt

Step 3.2: Interactive Content Calendar Generator

- User input workflow

Sample Output : TikTok Content Calendar for GlowNaturals

- Generated content calendar

Sample Output 2: LinkedIn Content Calendar for TechStart Academy

- Generated content calendar

Next Sample Output 3: